Marco Notaro

Curriculum Vitae

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Education

Oct 2015-Feb 2019 **Ph.D. in Computer Science**, *Thesis Title: "Hierarchical Ensemble Methods for Ontology-based Predictions in Computational Biology"*, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re.

Oct 2012-Feb 2015 **M.Sc. in Molecular Biotechnology and Bioinformatics (Curriculum Bioinformatics)**, *Thesis Title: "Hierarchical Ensemble Methods for Biological Ontologies Structured as Directed Acyclic Graphs"*, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re – Final Grade: 110/110.

Oct 2009-Oct 2012 **B.Sc. in Industrial and Environmental Biotechnology (Curriculum Bioinformatics)**, *Thesis Title: "High Resolution Maps of Methylation and Acetylation of Histones in the Human Genome"*, Milan University, Bioscience Department, Advisor: Prof. Giulio Pavesi – Final Grade: 101/110.

Research and Work Experiences

Feb 2019-Present **Scientific Collaborator**, at AnacletoLab - Computational Biology and Bioinformatics Laboratory, Milan University, Computer Science Department.

Feb 2019-Oct 2019 **Scientific Collaborator**, in the project "Data Model of Next-Generation DataBase Management System (DBMS)", Milan University, Computer Science Department. Supervisor: Prof. Marco Mesiti.

Oct 2015-Feb 2019 **PhD Research Activity**, at Computational Biology and Bioinformatics Group, Milan University, Computer Science Department, under the supervision of Prof. Giorgio Valentini.

Mar 2018-May 2018 **Visiting PhD Student**, at CITIC-UGR, Centro de Investigación en Tecnologías de la Información y de las Cominucaciones, Granada University, Spain, under the supervision of Dr. Carlos Cano Gutierrez.

Sep 2016-Jan 2017 **CAFA3 Challenge**, international challenge for the prediction of protein function and for the prediction of genes associated with abnormal phenotypes.

June 2016-Aug 2016 **Visiting PhD Student**, at Charité - Universitätsmedizin Berlin, Institute for Human and Medical Genetics, joining in the *Computational Biology Group* under the supervision of Prof. Peter Robinson.

Feb 2014-Feb 2015 **Master Degree Internship**, at Computational Biology and Bioinformatics Group, Milan University, Computer Science Department, Advisor: Prof. Giorgio Valentini, Co-Advisor: Dr. Matteo Re.

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Teaching Activity

May 2019 **Lecturer**, for the lecture on "Hierarchical Ensemble Methods for Ontology-based Prediction in Computational Biology" for the course Bioinformatics, M.Sc. in Computer Science.

Oct 2017-July 2018 **Co-advisor**, for a Bioinformatics thesis in M.Sc. in Computer Science.

May 2017 **Lecturer**, for the lecture on "Hierarchical Ensemble Methods for Structured Prediction with Applications in Computational Biology" for the course Bioinformatics, M.Sc. in Computer Science.

Oct 2015-Jan 2016 **Practical Assistant**, for the course "Statistics and Data Analysis" (R Laboratory), Milan University, Computer Science Department, Advisor: Prof. Dario Malchiodi (link).

Mar 2015-May 2015 **Practical Assistant**, for the course "Data Analysis" (R Laboratory), Milan University, Computer Science Department, Advisor: Anna Mari Zanaboni (link).

Summer Schools and International Meeting

12-16 Sep 2016 **Big Data for Precision Medicine and The 4th i2b2 European Meeting**, Pavia, IT.

Speaker at International Conferences and Poster Presentation

7-9 Sep 2016 **CIBB2017**, Cagliari, IT, took part as speaker by presenting the following works 1. Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions; 2. Disease Genes must Guide Data Source Integration in the Gene Prioritization Process.

9-12 Sep 2019 **Basel Life**, Basel, CH, poster presentation, title: Hierarchical ensemble methods for ontology-based predictions in computational biology and precision medicine.

Language Skills and Certifications

Italian Mother Tongue.

English Upper Intermediate Level, (Certified by Shenker Method -2013-).

Programming Languages

Main Programming Languages • **R**: machine learning; statistics and data analysis; data manipulation and data visualization (ggplot2) • **Perl**: parsing and analysis of raw text files; pattern matching by regular expression • **Bash**: scheduling computational experiments and executing commands in Linux environment

Side Programming Languages **JavaScript**: misc. and basic knowledge for maintaining my website • **Python**, **MATLAB** and **C++**: misc. and basic knowledge for scientific purpose

Markup Languages • **HTML** and **CSS** for maintaining my website • **LaTex** for writing scientific papers and reports

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Software Libraries

HEMDAG an R package: • implements several Hierarchical Ensemble Methods (HEMs) for Directed Acyclic Graphs (DAGs); • reconciles flat predictions with the topology of the ontology; • can enhance the predictions of virtually any flat learning methods by taking into account the hierarchical relationships between ontology classes; • guarantees biologically meaningful predictions that obey the true-path-rule, that is the biological and logical rule that governs the internal coherence of biomedical ontologies; • is specifically designed for exploiting the hierarchical relationships of DAG-structured taxonomies, such as the Human Phenotype Ontology (HPO) or the Gene Ontology (GO), but can be safely applied to tree-structured taxonomies as well (e.g. FunCat), since trees are DAGs; • scales nicely both in terms of the complexity of the taxonomy and in the cardinality of the examples; • provides several utility functions to process and analyze graphs; • provides several performance metrics to evaluate HEMs algorithms. HEMDAG is available both on github, biconda and cran repository alongside a step-by-step tutorial (link) showing how to apply HEMDAG to real case studies in Computational and Precision Medicine

obogaf::parser a Perl5 module to parse obo (Open Biological and Biomedical Ontology) and gaf (Gene Association File) file. obogaf::parser is available on github, bioconda and cpan repository alongside a step-by-step tutorial (link) showing how to apply obogaf::parser to bio-medical case studies

Research Interests

Thanks to high-throughput biotechnologies, we are assisting to a massive stacking of genomic, transcriptomic, proteomic and metabolomic data. Molecular biology is nowadays turning into a "data-rich" science. This unprecedented richness of raw data is meaningless until we cannot extract the interesting and useful "needles" of information from the big "haystack" of available data. This allow to us to have at the same time both a "bird's eye" and a "magnified" overview on a biological sample from a wide range of medically relevant disorders. "*The answer is there, blowing in the bulk*". However, this huge amount of raw data is of little use without groundbreaking bioinformatics and biostatistics methodologies able to disentangle, process, analyze and interpret them. In my research, I would like to strike a balance between two ways of doing science. On one hand I would like to focus on the design, development and application of novel statistical and machine learning approaches for analyzing heterogeneous and high-dimensional biomedical data. On the other hand, I would like to package such approaches into modular, scalable and user-friendly software tools, in order to make them available to the scientific community at large.

My research interests touch the fields of **Bioinformatics**, **Computational Biology** and **Machine Learning**. More precisely, my research lines can be grouped as follow:

- BIO-MOLECULAR NETWORKS: analysis and construction of complex biomolecular networks by making use of database such as STRING, BioGRID and UniProt;
- FLAT LEARNING METHODS: application of both supervised and semi-supervised learning methods to make "flat" predictions, i.e. predictions that do not consider the hierarchical structure of the label space;
- OUTPUT STRUCTURED PREDICTIONS: design and software implementation of hierarchical ensemble methods (HEMs) for the prediction of protein function (an open fundamental challenge in molecular biology) and for the prediction of human gene-abnormal phenotype associations (a crucial step to discover novel genes associated to Mendelian diseases). The HEMs' predictions are consistent with the ontology predictions, because they always obey to the "true path rule", the logical rule that governs the internal coherence of the biological ontologies, such as the Gene Ontology (GO) and the Human Phenotype Ontology (HPO).

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- NEXT GENERATION SEQUENCING DATA ANALYSIS: genetic variants found in sequencing data (e.g. single nucleotide variants and INDELs), copy number variation analysis.

Awards

Prediction of Human Phenotype Ontology terms by means of Hierarchical Ensemble Methods: awarded by the International Medical Informatics Association (IMIA) as one of the five best "Knowledge Representation and Management" (KRM) papers of 2017 in the field of Medical Informatics (link)

Publications

International Peer-Reviewed Journals

5. **Notaro M**, Frasca M, Petrini A, Valentini G. HEMDAG: a scalable and flexible hierarchical ensemble-based tool to improve Gene Ontology term predictions; 2019. Submitted to Plos Computional Biology.

4. Zhou N, Jiang Y, . . . , **Notaro M**, . . . , Radivojac P, Friedberg I. The CAFA challenge reports improved protein function prediction and new functional annotations for hundreds of genes through experimental screens. Genome Biology. 2019. Accepted for publication.

3. Perlasca P, Frasca M, Ba CT, **Notaro M**, Petrini A, Casiraghi E, Grossi G, Gliozzo J, Valentini G, Mesiti M. UNIPred-Web: a web tool for the integration and visualization of biomolecular networks for protein function prediction. BMC Bioinformatics. 2019. 20(1):422:1–422:19. doi:10.1186/s12859-019-2959-2.

2. Frasca M, Grossi G, Gliozzo J, Mesiti M, **Notaro M**, Perlasca P, Petrini A, Valentini G. A GPU-based algorithm for fast node label learning in large and unbalanced biomolecular networks. BMC Bioinformatics. 2018. 19-S(10):269–280. doi:10.1186/s12859-018-2301-4.

1. **Notaro M**, Schubach M, Robinson PN, Valentini G. Prediction of Human Phenotype Ontology terms by means of hierarchical ensemble methods. BMC Bioinformatics. 2017. 18(1):449:1–449:18. doi:10.1186/s12859-017-1854-y.

Proceedings of International Conferences and Peer-Reviewed Book Chapters

4. **Notaro M**, Schubach M, Frasca M, Mesiti M, Robinson PN, Valentini G. Ensembling Descendant Term Classifiers to Improve Gene - Abnormal Phenotype Predictions. In: Bartoletti M, Barla A, Bracciali A, Klau GW, Peterson L, Policriti A, Tagliaferri R, editors. Computational Intelligence Methods for Bioinformatics and Biostatistics. Cham: Springer International Publishing; 2019. doi:10.1007/978-3-030-14160-8_8. Available from: https://doi.org/10.1007/978-3-030-14160-8_8.

3. Frasca M, Fontaine JF, Valentini G, Mesiti M, **Notaro M**, Malchiodi D, Andrade-Navarro MA. Disease–Genes Must Guide Data Source Integration in the Gene Prioritization Process. In: Bartoletti M, Barla A, Bracciali A, Klau GW, Peterson L, Policriti A, Tagliaferri R, editors. Computational Intelligence Methods for Bioinformatics and Biostatistics. Cham: Springer International Publishing; 2019. doi:10.1007/978-3-030-14160-8_7. Available from: https://doi.org/10.1007/978-3-030-14160-8_7.

2. Robinson PN, Frasca M, Köhler S, **Notaro M**, Re M, Valentini G. A Hierarchical Ensemble Method for DAG-Structured Taxonomies. In: Schwenker F, Roli F, Kittler J, editors. Multiple Classifier Systems. Cham: Springer International Publishing; 2015. doi:10.1007/978-3-319-20248-8_2. Available from: https://doi.org/10.1007/978-3-319-20248-8_2.

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1. Valentini G, Köhler S, Re M, **Notaro M**, Robinson PN. Prediction of Human Gene - Phenotype Associations by Exploiting the Hierarchical Structure of the Human Phenotype Ontology. In: Ortuño F, Rojas I, editors. Bioinformatics and Biomedical Engineering. Cham: Springer International Publishing; 2015. doi:10.1007/978-3-319-16483-0_7. Available from: https://doi.org/10.1007/978-3-319-16483-0_7.

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